

MEANS TOOLBOX

t procedures • new 4.1–4.10 • old 7.1–7.10

ZONE A · ONE MEAN, PAIRED DIFFERENCES, TWO MEANS

ONE MEAN

$$s_{\bar{x}} = \frac{s}{\sqrt{n}}$$

$$\bar{x} \pm t^* \frac{s}{\sqrt{n}} \quad df = n - 1$$

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \quad df = n - 1$$

$H_0 : \mu = \mu_0$. Standard error $s_{\bar{x}}$ estimates $\sigma_{\bar{x}} = \sigma/\sqrt{n}$ using sample SD s .

TWO MEANS · INDEPENDENT GROUPS

$$s_{\bar{x}_1 - \bar{x}_2} = \sqrt{s_1^2/n_1 + s_2^2/n_2}$$

Left: SE estimates the sampling SD. Right: true SD when population SDs σ_1, σ_2 are known.

$$(\bar{x}_1 - \bar{x}_2) \pm t^* \sqrt{s_1^2/n_1 + s_2^2/n_2}$$

df: calculator, or $\min(n_1 - 1, n_2 - 1)$

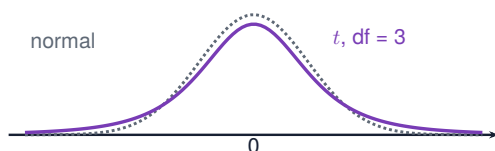
$$\sigma_{\bar{x}_1 - \bar{x}_2} = \sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}$$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}$$

df: calculator, or $\min(n_1 - 1, n_2 - 1)$

Test above assumes $H_0 : \mu_1 = \mu_2$. Calculator df is Welch's approximation; the minimum is conservative.

ZONE B · WHY t? + CONDITIONS IN ORDER



t because we replaced σ with s .

Heavier tails; more df $\rightarrow$ closer to normal.

- 1 • **Random:** random sample or randomized experiment.
- 2 • **10%:** $n < 0.10N$ when sampling without replacement.
- 3 • **Normal / Large Sample:** normal population, $n \geq 30$, or graph shows no strong skew or outliers.

Paired: check the **differences**.

Two samples: independent groups; check **each group**.

ZONE C · SAY IT IN WORDS

"We are ____% confident the true mean _____ is between _____ and _____ [units]."

For paired data, name the mean difference μ_d ; for independent groups, name $\mu_1 - \mu_2$.

ZONE D · WATCH OUT

Paired or two-sample? Let the study design decide. Repeated measurements or natural pairs use differences; separate independent groups use two-sample t.

Keep group variances separate for two-sample t. Use the unpooled SE above.